

# Linux Class 9 - Practice Task

Practising gzip, tar and the sed replace options

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This is the practice I did after class 9. I made a folder called practice9.

## The files I made first

I made a small file called test.txt with 8 lines. Every line has the word is in it three times. I wrote it that way on purpose, because the notes talk about replacing the 1st or the 2nd occurrence, and with three on every line I can actually see which one changed.

The lines are about different things so it does not get boring: rahul, delhi, mango, linux, cricket, sunday, maths and train.

Then I needed some bigger files for gzip, because a tiny file does not compress much. So I joined test.txt to itself five times with cat and made file1.txt, and copied that to file2.txt and file3.txt:

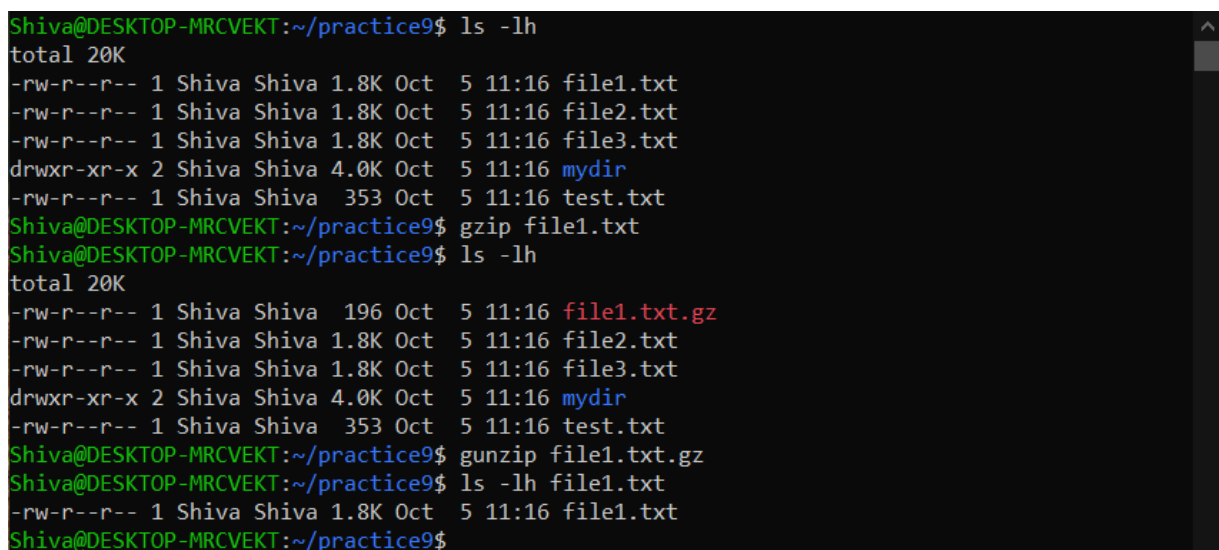
```
cat test.txt test.txt test.txt test.txt test.txt > file1.txt
cp file1.txt file2.txt
cp file1.txt file3.txt
```

I also made an empty folder called mydir, to try the note that says gzip does not work on folders.

## gzip and gunzip

gzip makes a file smaller and gunzip puts it back. I ran ls -lh before and after so the size is visible.

```
ls -lh
gzip file1.txt
ls -lh
gunzip file1.txt.gz
```



```
Shiva@DESKTOP-MRCVEKT:~/practice9$ ls -lh
total 20K
-rw-r--r-- 1 Shiva Shiva 1.8K Oct  5 11:16 file1.txt
-rw-r--r-- 1 Shiva Shiva 1.8K Oct  5 11:16 file2.txt
-rw-r--r-- 1 Shiva Shiva 1.8K Oct  5 11:16 file3.txt
drwxr-xr-x 2 Shiva Shiva 4.0K Oct  5 11:16 mydir
-rw-r--r-- 1 Shiva Shiva 353 Oct  5 11:16 test.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$ gzip file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$ ls -lh
total 20K
-rw-r--r-- 1 Shiva Shiva 196 Oct  5 11:16 file1.txt.gz
-rw-r--r-- 1 Shiva Shiva 1.8K Oct  5 11:16 file2.txt
-rw-r--r-- 1 Shiva Shiva 1.8K Oct  5 11:16 file3.txt
drwxr-xr-x 2 Shiva Shiva 4.0K Oct  5 11:16 mydir
-rw-r--r-- 1 Shiva Shiva 353 Oct  5 11:16 test.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$ gunzip file1.txt.gz
Shiva@DESKTOP-MRCVEKT:~/practice9$ ls -lh file1.txt
-rw-r--r-- 1 Shiva Shiva 1.8K Oct  5 11:16 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$
```

Screenshot 1: file1.txt is 1.8K, after gzip it is 196 bytes, and gunzip brings it back to 1.8K.

The part I did not expect is that file1.txt disappeared. gzip does not make a copy, it replaces the file with file1.txt.gz. gunzip does the same in reverse and removes the .gz. So after gzip there is no file1.txt any more until I unzip it.

1.8K went down to 196 bytes, which is very small. That is because my file is the same 8 lines repeated five times, and gzip is good at repeated text.

## Many files at once, and the one thing gzip will not do

gzip takes more than one file name:

```
gzip file1.txt file2.txt file3.txt
```

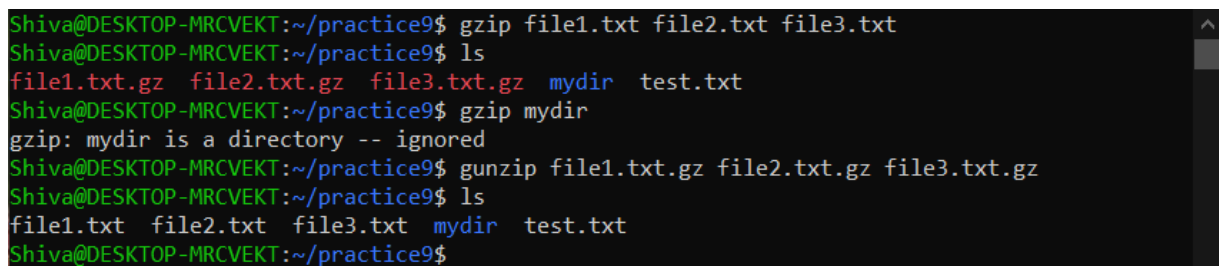
This did not join them. It made three separate files, file1.txt.gz, file2.txt.gz and file3.txt.gz. That is the difference between gzip and tar which I did next.

Then I tried it on the folder, like the note says:

```
gzip mydir
```

It refused and printed gzip: mydir is a directory -- ignored. So the note is right, gzip works on files only. For a folder I have to use tar.

```
gunzip file1.txt.gz file2.txt.gz file3.txt.gz
```



```
Shiva@DESKTOP-MRCVEKT:~/practice9$ gzip file1.txt file2.txt file3.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$ ls
file1.txt.gz file2.txt.gz file3.txt.gz mydir test.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$ gzip mydir
gzip: mydir is a directory -- ignored
Shiva@DESKTOP-MRCVEKT:~/practice9$ gunzip file1.txt.gz file2.txt.gz file3.txt.gz
Shiva@DESKTOP-MRCVEKT:~/practice9$ ls
file1.txt file2.txt file3.txt mydir test.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$
```

Screenshot 2: three files zipped, the error on the folder, and all three unzipped again.

## tar - putting many files into one

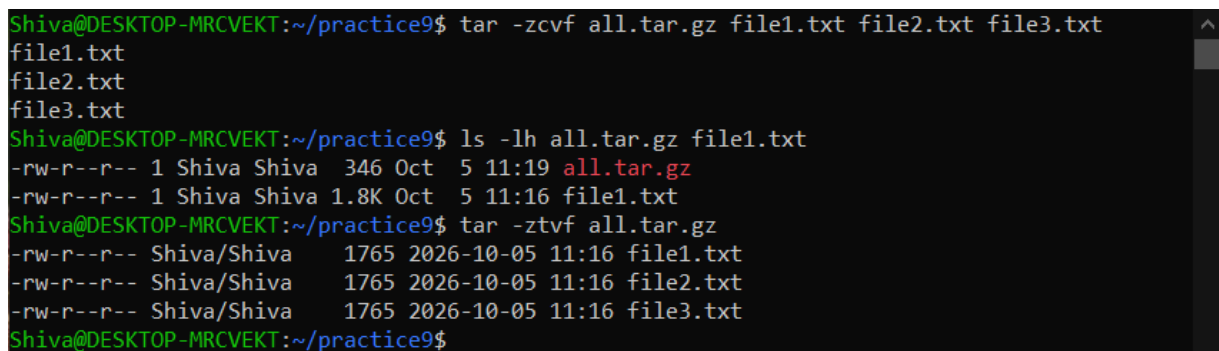
tar joins files together. On its own it does not make them smaller, but with -z it also runs gzip, so I get one file that is both joined and compressed.

```
tar -zcvf all.tar.gz file1.txt file2.txt file3.txt
```

The -v made it print the three names as it added them. Then I checked the size and looked inside without opening it:

```
ls -lh all.tar.gz file1.txt
```

```
tar -ztvf all.tar.gz
```



```
Shiva@DESKTOP-MRCVEKT:~/practice9$ tar -zcvf all.tar.gz file1.txt file2.txt file3.txt
file1.txt
file2.txt
file3.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$ ls -lh all.tar.gz file1.txt
-rw-r--r-- 1 Shiva Shiva 346 Oct  5 11:19 all.tar.gz
-rw-r--r-- 1 Shiva Shiva 1.8K Oct  5 11:16 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$ tar -ztvf all.tar.gz
-rw-r--r-- Shiva/Shiva 1765 2026-10-05 11:16 file1.txt
-rw-r--r-- Shiva/Shiva 1765 2026-10-05 11:16 file2.txt
-rw-r--r-- Shiva/Shiva 1765 2026-10-05 11:16 file3.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$
```

Screenshot 3: making the archive, its size, and the list of what is inside it.

all.tar.gz is 346 bytes while one of the files on its own is 1.8K. So three files of 1.8K each went into one file of 346 bytes. The -t listing shows the three names with their real sizes, 1765 bytes each, and it did not unpack anything.

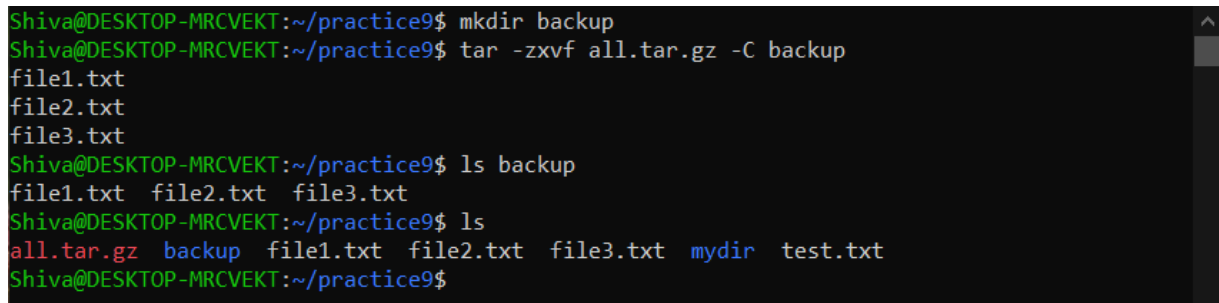
The order of the letters matters for f. The archive name has to come straight after the f, because that is what f means. I wrote -zcvf and then all.tar.gz, and it worked.

| Letter | What it means                                                        |
|--------|----------------------------------------------------------------------|
| -c     | Create a new archive.                                                |
| -x     | Extract an archive that already exists.                              |
| -t     | Only list what is inside, without taking it out.                     |
| -z     | Also compress it with gzip. This is why the name ends in .tar.gz.    |
| -v     | Verbose. Print the file names on the screen as it works.             |
| -f     | The next word is the archive name. This letter must be the last one. |

## Taking the archive apart again

-x is extract. I made a folder called backup first and used -C so the files would come out there instead of on top of the ones I already have:

```
mkdir backup
tar -zxvf all.tar.gz -C backup
ls backup
```



```
Shiva@DESKTOP-MRCVEKT:~/practice9$ mkdir backup
Shiva@DESKTOP-MRCVEKT:~/practice9$ tar -zxvf all.tar.gz -C backup
file1.txt
file2.txt
file3.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$ ls backup
file1.txt file2.txt file3.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$ ls
all.tar.gz backup file1.txt file2.txt file3.txt mydir test.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$
```

Screenshot 4: extracting into the backup folder, and the three files are there.

The last ls shows my folder now has all.tar.gz, the backup folder, the three files, mydir and test.txt. Nothing was lost.

## sed - which occurrence gets replaced

Now the sed part. I am replacing the word is with was. I printed the file with cat -n first so the line numbers are there to check against.

```
sed 's/is/was/' test.txt
sed 's/is/was/2' test.txt
```

```

Shiva@DESKTOP-MRCVEKT:~/practice9$ cat -n test.txt
 1 rahul is tall rahul is kind rahul is smart
 2 delhi is big delhi is old delhi is busy
 3 mango is sweet mango is yellow mango is juicy
 4 linux is free linux is fast linux is open
 5 cricket is fun cricket is long cricket is famous
 6 sunday is free sunday is late sunday is lazy
 7 maths is hard maths is useful maths is logic
 8 train is slow train is cheap train is safe
Shiva@DESKTOP-MRCVEKT:~/practice9$ sed 's/is/was/' test.txt
rahul was tall rahul is kind rahul is smart
delhi was big delhi is old delhi is busy
mango was sweet mango is yellow mango is juicy
linux was free linux is fast linux is open
cricket was fun cricket is long cricket is famous
sunday was free sunday is late sunday is lazy
maths was hard maths is useful maths is logic
train was slow train is cheap train is safe
Shiva@DESKTOP-MRCVEKT:~/practice9$ sed 's/is/was/2' test.txt
rahul is tall rahul was kind rahul is smart
delhi is big delhi was old delhi is busy
mango is sweet mango was yellow mango is juicy
linux is free linux was fast linux is open
cricket is fun cricket was long cricket is famous
sunday is free sunday was late sunday is lazy
maths is hard maths was useful maths is logic
train is slow train was cheap train is safe
Shiva@DESKTOP-MRCVEKT:~/practice9$

```

Screenshot 5: the file, then only the 1st is on each line changed, then only the 2nd.

The first command changed only the first is on every line. The notes say `sed 's/is/was/1'` does exactly the same, and that is true, the 1 is just saying what sed already does.

With 2 it skipped the first one and changed the second one on every line. This is per line, not per file. Every line got one change, not just line 2.

## All of them, and saving the change

`g` means all of them on the line:

```
sed 's/is/was/g' test.txt
```

That changed all three on every line. Still nothing was saved, it only printed on the screen. `test.txt` was exactly the same afterwards.

To actually save it I need `-i`. I made a copy first so I would not spoil `test.txt`, because `-i` has no undo:

```

cp test.txt copy.txt
sed -i '2s/is/was/3' copy.txt
cat -n copy.txt

```

This one is from the notes: line 2, 3rd occurrence, and save it. `sed -i` printed nothing at all, which made me think it failed, but when I opened `copy.txt` line 2 was `delhi is big delhi is old delhi was busy`. Only the third is changed and only on line 2. So it worked and the silence is normal, because the output went into the file.

```

Shiva@DESKTOP-MRCVEKT:~/practice9$ sed 's/is/was/g' test.txt
rahul was tall rahul was kind rahul was smart
delhi was big delhi was old delhi was busy
mango was sweet mango was yellow mango was juicy
linux was free linux was fast linux was open
cricket was fun cricket was long cricket was famous
sunday was free sunday was late sunday was lazy
maths was hard maths was useful maths was logic
train was slow train was cheap train was safe
Shiva@DESKTOP-MRCVEKT:~/practice9$ cp test.txt copy.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$ sed -i '2s/is/was/3' copy.txt
Shiva@DESKTOP-MRCVEKT:~/practice9$ cat -n copy.txt
 1 rahul is tall rahul is kind rahul is smart
 2 delhi is big delhi is old delhi was busy
 3 mango is sweet mango is yellow mango is juicy
 4 linux is free linux is fast linux is open
 5 cricket is fun cricket is long cricket is famous
 6 sunday is free sunday is late sunday is lazy
 7 maths is hard maths is useful maths is logic
 8 train is slow train is cheap train is safe
Shiva@DESKTOP-MRCVEKT:~/practice9$

```

Screenshot 6: g changes every one, then -i saves one change on line 2 into copy.txt.

## Picking a line, and picking a range

A number in front of the s is a line number. A number at the end is an occurrence. I keep mixing these two up, so I did all three forms one after the other.

```
sed '5s/is/was/2' test.txt
```

```
sed '2,6s/is/was/g' test.txt
```

```
sed '6,$s/is/was/g' test.txt
```

```

Shiva@DESKTOP-MRCVEKT:~/practice9$ sed '5s/is/was/2' test.txt
rahul is tall rahul is kind rahul is smart
delhi is big delhi is old delhi is busy
mango is sweet mango is yellow mango is juicy
linux is free linux is fast linux is open
cricket is fun cricket was long cricket is famous
sunday is free sunday is late sunday is lazy
maths is hard maths is useful maths is logic
train is slow train is cheap train is safe
Shiva@DESKTOP-MRCVEKT:~/practice9$ sed '2,6s/is/was/g' test.txt
rahul is tall rahul is kind rahul is smart
delhi was big delhi was old delhi was busy
mango was sweet mango was yellow mango was juicy
linux was free linux was fast linux was open
cricket was fun cricket was long cricket was famous
sunday was free sunday was late sunday was lazy
maths is hard maths is useful maths is logic
train is slow train is cheap train is safe
Shiva@DESKTOP-MRCVEKT:~/practice9$ sed '6,$s/is/was/g' test.txt
rahul is tall rahul is kind rahul is smart
delhi is big delhi is old delhi is busy
mango is sweet mango is yellow mango is juicy
linux is free linux is fast linux is open
cricket is fun cricket is long cricket is famous
sunday was free sunday was late sunday was lazy
maths was hard maths was useful maths was logic
train was slow train was cheap train was safe
Shiva@DESKTOP-MRCVEKT:~/practice9$

```

Screenshot 7: one line and one occurrence, then a range of lines, then line 6 to the end.

The first one touched only line 5, cricket, and on that line only the second is. Everything else stayed the same.

The second one is a range. Lines 2 to 6 were fully changed and lines 1, 7 and 8 were left alone.

The third one uses \$, which means the last line. So it changed line 6 down to the end, and I did not have to know that my file has 8 lines. That is useful on a big file.

| Command                       | What it changed                                |
|-------------------------------|------------------------------------------------|
| sed 's/is/was/' test.txt      | Only the 1st is on each line.                  |
| sed 's/is/was/1' test.txt     | The same thing. 1 is what sed does anyway.     |
| sed 's/is/was/2' test.txt     | Only the 2nd is on each line.                  |
| sed 's/is/was/g' test.txt     | Every is on every line.                        |
| sed '5s/is/was/2' test.txt    | Line 5 only, and on that line only the 2nd is. |
| sed -i '2s/is/was/3' copy.txt | Line 2, 3rd is, and saved into the file.       |
| sed '2,6s/is/was/g' test.txt  | Every is on lines 2 to 6.                      |
| sed '6,\$s/is/was/g' test.txt | Every is from line 6 to the last line.         |